Verification and Validation (VV)

VV Overview

Required PA Information

Intent

Confirms selected solutions and components meet their requirements, and demonstrates selected solutions and components fulfill their intended use in their target environment.

Value

Increases the likelihood that the solution will satisfy the customer.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

VV 1.1	Perform verification to ensure the requirements are implemented and record and communicate results.
VV 1.2	Perform validation to ensure the solution will function as intended in its target environment and record and communicate results.

Level 2

VV 2.1	Select components and methods for verification and validation.
VV 2.2	Develop, keep updated, and use the environment needed to support verification and validation.
VV 2.3	Develop, keep updated, and follow procedures for verification and validation.

Level 3

VV 3.1	Develop, keep updated, and use criteria for verification and validation.
VV 3.2	Analyze and communicate verification and validation results.

Additional PA Explanatory Information

Verification and validation activities include identifying corrective actions from verification and validation activities and tracking them to closure.

Verification and validation address different perspectives:

- Verification addresses whether the work product and solution properly reflect the specified requirements, i.e., "you are building it right."
- Validation demonstrates that the solution will fulfill its intended use in its target environment, i.e., "you are building the right thing."

Verification and validation activities typically include incremental, iterative, or concurrent processes that:

- Begin with requirements
- Continue through developing work products and the completed solution
- End with transitions to operations and sustainment

Validation activities can be applied to all aspects of the solution in any of its target environments, including:

- Development
- Testing
- Operations
- Training
- Manufacturing
- Maintenance
- Support

Validation activities use approaches like verification, e.g., test, analysis, inspection, demonstration, simulation. Typically, end users and other affected stakeholders are involved in the project's validation activities.

Related Practice Areas

Requirements Development and Management (RDM)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Agile teams typically define a definition of done for each user story or requirement. Work is performed until the definition of done is met for each user story. Acceptance from the product owner is obtained during the Sprint review using defined acceptance criteria.

An agile team considers testing and demonstrations to address how the user will operate the solution in the intended environment. Recorded results show the status of verification and validation activities and provide an opportunity for analysis. Refer to Table VV-1: Example Agile Verification and Validation Activities for examples of agile verification and validation activities.

Table VV-1: Example Agile Verification and Validation Activities

Agile Activities	Example Verification and Validation Activities
Code reviews	Verification: Includes self-reviews and peer reviews against coding standards and unit test coverage
Manual or automated testing	Verification: Targets and checks logic flow and identifies criteria to test that code meets requirements
Sprint reviews / demonstrations	Validation: Paves the way for user acceptance testing (UAT) and system testing, by providing interim demonstrations to confirm epics and stories meet customer expectations, and identifies any issues or gaps

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Ensure appropriate techniques are considered for verifying and validating safety requirements and recorded safety targets. Consider the following:

- Verification of components and the product using simulation
- Testing techniques requiring testing to a specified level of structural, data, or path coverage
- Use of field-use data, including reference sites and service histories
- Prototype testing
- Stress testing
- Accelerated service-life testing
- Test cases and scripts aligned to potential points of operational failure
- Use of mathematical models to validate high safety assurance cases, e.g., simulations to verify medical device regulations

These activities rely on other verification and validation to demonstrate that the product performs its specified function in the intended-use environment and that unintended functionality is not present.

If new hazards are discovered or known hazards are determined to have a higher risk category than previously assessed, ensure the risk is formally accepted by the approving authority.

If it is not possible to eliminate an identified hazard, then reduce the associated risk to a level that is deemed acceptable to the end user or approving authority.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization

Incorporate security considerations into all aspects of verification and validation activities, e.g., environment setup, test procedures, test cases, and installation checklists. Ensure that verification and validation environments and technologies meet defined security standards.

Plan for what security work products should be selected for verification and validation activities. For example, select the security requirements, designs, and prototypes that are the best predictors of how well the solution or solution components satisfies end user security needs. Perform verification and validation activities early, e.g., concept and exploration phases, and incrementally throughout the solution lifecycle, including through the transition to operations and retirement. Types of verification and validation activities to consider include:

- Penetration testing
- Fuzz testing
- Security focused peer reviews
- Tool-based security code reviews
- Static code analysis
- Dynamic analysis
- Security threat analysis reviews
- Simulations
- White, gray, and black box testing
- Review use of tags, cryptographic hash verifications, or digital signatures

During verification and validation activities, include security representatives to participate and ensure the security risks, threats, and vulnerabilities have been evaluated, and that security requirements are being met. Ensure management accepts any residual security risks and mitigations associated with the solution. If the risk or mitigation is deemed unacceptable, then the designed solution and associated technologies must be reworked.

When performing verification and validation activities, consider a dedicated test environment to control the overall security testing parameters. The environment should reflect the intended operational environment and if operational data is used during verification or validation, remove or sanitize sensitive information, e.g., Personal Identifiable Information (PII), Personal Health Information (PHI), and user-specific passwords. As changes are made during security verification and validation, track solutions tested and reviewed for security purposes and place them under an appropriate level of configuration control. Ensure compatibility among the interfaces or connections by considering how to integrate solutions with the management of internal and external interfaces or connections of secure solutions and solution components.

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Approaches and processes for verification and validation include:

- Each phase of the service operations, e.g., preparation, operation and monitoring, maintenance, enhancements, retirement
- A clear description of service environments including service development and setup, operational service approaches and objectives
- Service system components and processes
- Development, staging, and production environments to pilot or prototype the service operation
- Different approaches, systems, and processes used to verify and validate service delivery and operation

Level 1

VV 1.1

Required Practice Information

Practice Statement

Perform verification to ensure the requirements are implemented and record and communicate results.

Value

Reduces the cost of addressing requirements issues and increases customer satisfaction.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Verification throughout the lifecycle helps to ensure that requirements are implemented correctly and issues are identified early.

Example Activities

Example Activities	Further Explanation
Perform the verification of selected work products and solutions against their requirements.	
Record and communicate the results of verification activities.	
Identify action items resulting from verification.	

Example Work Products

Example Work Products	Further Explanation
Verification results	
Action items	

VV 1.2

Required Practice Information

Practice Statement

Perform validation to ensure the solution will function as intended in its target environment and record and communicate results.

Value

Increases the likelihood that the result provides the right solution to meet customer expectations.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

This section left blank for future content.

Example Activities

Example Activities	Further Explanation
Validate selected work products and solutions with stakeholders throughout the lifecycle to ensure they function as intended in their target environment.	
Analyze and communicate the results of validation activities.	

Example Work Products

Example Work Products	Further Explanation
Validation results	
Analysis results	

Level 2

VV 2.1

Required Practice Information

Practice Statement

Select components and methods for verification and validation.

Value

Produces solutions that meet or exceed customer expectations and needs.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Selecting work products for verification and validation enables early:

- Identification and resolution of defects
- Requirements refinement
- Indication of whether the proposed solution will work in the target environment
- Assessment of the proposed solution's fitness for use
- Confidence that the proposed solution will meet customers' needs and expectations
- Understanding of the solution among stakeholders

Verification and validation activities may result in derived requirements. Ensure the derived requirements are included in the requirements set.

Example Activities

Example Activities	Further Explanation
Select solution components for verification and validation.	Select solution components based on their contribution to meeting objectives for the solution. For each solution component, determine the: • Scope of the verification and validation activities • Requirements to be satisfied • Methods or tools to be used Solution components that can be verified and validated include: • Requirements, designs, and constraints • Acquired and developed solutions and related components • User interfaces or connections

Example Activities	Further Explanation
	• User and operational manuals • Training materials • Process documentation
Identify requirements to be satisfied by each selected work product.	When identifying requirements for each selected work product, consult the traceability matrix (or other requirements traceability information) updated as part of managing requirements for the work.
Determine which customer requirements and end user needs will be validated.	Target environments may include: • Testing • Operations • Maintenance • Training • Support services
Define, record, and keep updated verification and validation methods to be used for each selected solution.	Examples of verification methods include: • Requirements inspection • Demonstration • Load, stress, and performance testing • Functional, interface or connection, and integration testing • Prototyping, modeling, and simulation Examples of validation methods include: • Reviews with end users • Prototype demonstrations • Functional demonstrations • Pilots • Tests of solution components by end users and other affected stakeholders Attributes to consider for verification and validation may include: • Quality • Functionality • Maintainability • Reliability
Review the validation selection, roles, responsibilities, constraints, and methods with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Lists of solution components selected for verification and validation	

Example Work Products	Further Explanation
Verification and validation methods for each selected solution component	
List of requirements to be verified and validated	

Related Practice Areas

Decision Analysis and Resolution (DAR)

Requirements Development and Management (RDM)

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Verification and validation for hardware engineering typically considers:

- Environmental conditions, e.g., pressure, temperature, vibration, humidity
- Input ranges, e.g., input power could be rated at 20V to 32V for a planned nominal of 28V
- Variations induced from part-to-part tolerance issues

Hardware verification normally tests most variables separately except when problematic interactions are suspected. Hardware verification and validation methods include:

- Modeling to validate form, fit, and function of mechanical designs
- Thermal modeling
- Maintainability and reliability analysis
- Timeline demonstrations
- Simulations

Verification and validation for software engineering typically includes:

- Simulation
- Traceability studies
- Functional reviews or audits
- Physical reviews or audits
- Peer reviews
- Demonstrations
- Prototypes
- Formal reviews
- Module, regression, and system integration testing

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Service system components selected for verification and validation should cover:

- Services delivered in accordance with service delivery approaches and agreements
- Changes managed for the service delivery system
- Receipt and processing of service requests
- Sustainment of service delivery performance when changes occur
- Service agreements that describe what a service provider will deliver to the customer, and includes:
  - Service level and availability targets
  - Responsibilities of the service provider, customer, and end user based on their process role
  - Communications channels and feedback mechanisms

Methods used for verification and validation of service system components may include:

- Prototyping
- Piloting
- Service delivery walk throughs
- Peer review
- User acceptance testing

VV 2.2

Required Practice Information

Practice Statement

Develop, keep updated, and use the environment needed to support verification and validation.

Value

Minimizes project delays by ensuring that verification and validation environments are ready when needed.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

The requirements for verification and validation environments are driven by the:

- Selected solution or solution components
- Type of work products, e.g., design, prototype, final version
- Methods or tools used for verification and validation
- Physical constraints, e.g., temperature, pressure, humidity

Elements in a verification or validation environment include:

- Test tools that work with the solution or solution components being verified or validated
- Embedded test equipment or software
- Simulated subsystems or components and their interfaces or connections
- Interfaces or connections to the operational environment
- Facilities and customer supplied products
- End users and operators
- Scenarios
- Actual or targeted physical environments, e.g., weather conditions, space, vacuum

Using these requirements and elements, the verification and validation environments can be acquired, developed, reused, modified, or obtained depending on the needs of the solution.

Example Activities

Example Activities	Further Explanation
Identify requirements for the verification and validation environment.	For solutions that include service delivery, requirements should be identified for all phases of services operation and delivery.
Identify customer-supplied solutions and components.	This is typically done in the validation environment if the customer has existing facilities or components that are part of the intended environment.
Identify verification and validation resources, equipment, and tools.	Identify verification and validation resources that are available for reuse and modification.
Develop or acquire and keep the verification and validation environments updated.	

Example Work Products

Example Work Products	Further Explanation
Verification environment	
Validation environment	

VV 2.3

Required Practice Information

Practice Statement

Develop, keep updated, and follow procedures for verification and validation.

Value

Reduces costs for performing the activities and strengthens more predictable performance.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Verification procedures ensure that work products meet their requirements. Validation procedures ensure that the solution or solution component will fulfill its intended use when placed in its target environment.

Example Activities

Example Activities	Further Explanation
Identify criteria for selecting work products and activities for verification and validation.	
Develop and keep updated procedures for verification and validation.	Procedures may cover: • Maintenance • Set up and support of test and evaluation facilities • Training • Management and acceptable use of test data
Perform verification and validation in accordance with the procedures.	

Example Work Products

Example Work Products	Further Explanation
Verification procedures	
Validation procedures	
Verification and validation results	

Level 3

VV 3.1

Required Practice Information

Practice Statement

Develop, keep updated, and use criteria for verification and validation.

Value

Minimizes waste by ensuring the verification and validation activities focus on critical needs.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Examples of sources for verification and validation criteria include:

- Solution or solution component requirements
- Business process descriptions
- Standards, regulatory and legal requirements
- Customer requirements
- Customer acceptance criteria
- Solution performance
- Contracts and agreements

Example Activities

Example Activities	Further Explanation
Develop verification and validation criteria and refine them as work progresses.	Criteria may include: • Specification of test inputs and outputs • Description of expected results • Description of acceptable results • Target service levels and expected results • Service objectives and measurements • Customers types and expectations

Example Work Products

Example Work Products	Further Explanation
Verification criteria	
Validation criteria	

VV 3.2

Required Practice Information

Practice Statement

Analyze and communicate verification and validation results.

Value

Improves verification and validation effectiveness over time.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Analysis and communication of verification and validation results ensure that issues receive the appropriate attention from stakeholders and management.

Example Activities

Example Activities	Further Explanation
Compare actual results to expected results.	
Identify the results that do not meet established criteria for verification.	
Identify the results that do not meet established criteria for validation.	
Analyze verification and validation results that do not meet established criteria and determine corrective actions.	Results may include defect, error, or issue analysis.
Submit improvement proposals for the verification and validation process.	
Record and communicate analysis results and corrective actions to affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Results of actual to expected comparisons for verification and validation	
Analysis results	

Example Work Products	Further Explanation
Corrective actions	
Improvement proposals	